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Bihar Engineering University, Patna

B.Tech 7th Semester Exam., 2025

Code: 155703 ***Time: 03 Hours***

Semester: VII ***Subject: Application of IoT in Robotics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/ answer in the following(Any seven question only): **[2 x 7 = 14]**

(a) LoRa technology is mainly used for
- (i) High data rate communication
- (ii) Long-range low-power communication
- (iii) Short-range indoor networking
- (iv) Wired sensor networks

(b) Actuators convert
- (i) Electrical energy into motion
- (ii) Software into hardware
- (iii) Data into signals
- (iv) Heat into electricity

(c) Which protocol is lightweight and suitable for IoT robotics?
- (i) HTTP
- (ii) MQTT
- (iii) FTP
- (iv) SMTP

(d) Digital Twin refers to
- (i) Backup robot
- (ii) Robot clone
- (iii) Virtual replica of physical system
- (iv) Simulation software only

(e) Which communication technology is used for short range communication?
- (i) Wi-Fi
- (ii) LTE
- (iii) ZigBee
- (iv) Ethernet

(f) The process which combines data from multiple devices into a unified format at the IoT gateway is called
- (i) Data enrichment
- (ii) Device authentication
- (iii) Data consolidation
- (iv) Packet routing

(g) CSV data format is best described as
- (i) Binary structured data
- (ii) Comma-separated plain text data
- (iii) Hierarchical data format
- (iv) Encrypted data format

(h) Edge computing refers to
- (i) Processing data only in the cloud
- (ii) Storing data in remote servers
- (iii) Processing data near the data source
- (iv) Using only centralized systems

(i) Select the IoT platform for device telemetry and cloud integration provided by Microsoft.
- (i) Azure Functions
- (ii) Azure IoT Hub
- (iii) Azure DevOps
- (iv) Azure Storage

(j) MQTT follows which communication model?
- (i) Client–Server
- (ii) Publish–Subscribe
- (iii) Peer-to-Peer
- (iv) Master–Slave

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Define the Internet of Things. Explain its main components and describe the architecture of an IoT system with a neat block diagram.

(b) Explain how robotic perception is enhanced using IoT sensors. Discuss the role of multi-sensor data in improving environmental awareness and decision-making in robots.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Describe various types of IoT sensors and actuators used in robotic systems. Discuss the selection criteria of sensors and actuators for robotic systems.

(b) Explain the interfacing of IoT sensors and actuators with microcontrollers on the Raspberry Pi platform.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) How data enrichment and data consolidation functions are performed at IoT Gateway? Describe with suitable example.

(b) What are the internet connectivity options in IoT devices? Highlight the role of IPv6 in large-scale IoT deployments.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) How middleware helps in integrating IoT devices with robotic systems? Discuss how device abstraction and data management is done at middleware.

(b) Why is edge computing important in robotics? Discuss its advantages over cloud-only computing with suitable examples.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain the process of data acquisition from IoT sensors and actuators in robotic systems. Describe various data logging formats used in IoT-based robotic systems.

(b) What are data processing techniques in IoT systems? Explain filtering, aggregation and normalization techniques used for processing sensor data in robotics with suitable examples.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Describe autonomous delivery robots. Explain how IoT technologies like sensors, GPS, cloud connectivity and real-time data analysis enable autonomous navigation and delivery.

(b) What is Industrial IoT? Explain various IIoT-based robotics applications in manufacturing industries such as automation, predictive maintenance and smart production lines.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) How M2M system layers ensure design standardization in IoT systems to support interoperability among heterogeneous devices?

(b) Discuss the case study of IoT-enabled drones. Explain their architecture and various applications in surveillance, agriculture and disaster management.

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Types of robots

(b) IoT in Healthcare

(c) Web communication protocols

(d) Google Cloud IoT

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